KHALED ALBAKOUR

Geospatial & Precision Agriculture Developer

GIS • Satellite Remote Sensing • Google Earth Engine • Agricultural Decision Support

Hama, Syria | +963 958 687 748 | khaled.albakour.geo@gmail.com
Portfolio: AGRI-EYE website | GitHub: Khaled8M | AGRI-EYE organization

PROFESSIONAL PROFILE

Self-taught geospatial developer focused on GIS, satellite remote sensing and agricultural decision-support workflows. Developed Google Earth Engine JavaScript tools for wheat monitoring and relative soil-condition screening, and built AGRI-EYE TRIO K, a hybrid desktop platform combining online satellite workflows with offline geospatial analysis. Practical experience with multi-source satellite and environmental datasets, field boundaries, spatial sampling, dashboards and GIS exports, supported by six years of hands-on IT service experience.

Target roles: Junior GIS Analyst • GIS Technician • Remote Sensing Analyst • Precision Agriculture / AgTech • Geospatial Software Support

CORE TECHNICAL COMPETENCIES

GIS & Earth observation: Google Earth Engine; Sentinel-1/Sentinel-2; Landsat 8; MODIS; ERA5-Land; GLDAS; CHIRPS; SRTM

Programming & web: JavaScript (Earth Engine API); practical Python application scripting/integration; HTML/CSS/JavaScript; debugging and defensive data handling

Analysis: NDVI, EVI, MSAVI, NDWI, GNDVI, SAVI, NDRE; crop-growth mapping; relative soil-condition, moisture and erosion-risk screening; spatial sampling and area statistics

Data & delivery: GeoTIFF; KML/KMZ; GeoJSON; Shapefile; interactive charts/dashboards; PDF reporting; Git/GitHub; Cloudflare Pages; Netlify; Microsoft Excel (advanced)

SELECTED GEOSPATIAL PROJECTS

AGRI-EYE TRIO K - Hybrid Geospatial Agricultural Decision-Support Platform | *Founder / Independent Developer | 2024-present*

- Designed and tested a Windows desktop application combining online Google Earth Engine workflows with offline analysis of supported local geospatial and GeoTIFF data.
- Implemented farm-boundary mapping and crop workflows for wheat, potato, corn, olive and peanut/groundnut, alongside universal and soil-analysis workflows.
- Integrated vegetation and supported moisture/stress proxy indicators, spatial summaries, candidate zones for field inspection, PDF reports and supported geospatial exports including GeoTIFF, KML/KMZ, GeoJSON and Shapefile.
- Tested the V1.2.1 preparation with Italian interface localization and Italian-language PDF report workflow; indicators are treated as decision-support proxies, not standalone diagnoses or guaranteed yield measurements.

Agricultural Wheat Monitoring System | *Google Earth Engine JavaScript | 2023-present; major development in 2025 (v7.1)*

- Built an approximately 900-line GEE application integrating Sentinel-1, Sentinel-2, Landsat 8, MODIS, ERA5-Land, GLDAS, CHIRPS and SRTM.
- Implemented vegetation/environmental indices, Zadoks-based wheat growth-stage classification, crop growth-zone mapping, spatial sampling, area statistics, interactive charts and GeoTIFF export.
- Developed water-stress screening and a climate/spectral Yellow Rust risk-screening workflow, plus yield-potential estimation; these outputs are model-based indicators requiring independent field validation.
- Improved server-side reliability by addressing geometry/area edge cases, reducing redundant evaluate() calls and refining Sentinel-1 scaling and SWIR resampling.

KHALED ALBAKOUR | Geospatial & Precision Agriculture Developer

ADDITIONAL GEOSPATIAL WORK

Italian Durum Wheat Monitoring Workflow | *Google Earth Engine | 2026*

Prototype / ongoing development

- Developed/adapted a prototype Italy-oriented durum-wheat monitoring workflow, extending existing crop-monitoring scripts for seasonal vegetation tracking and geospatial reporting.
- Prototype status: continued QA, documentation and agronomic/field validation are still required before making performance claims.

Soil Analysis System Engine | *Google Earth Engine | Scientific core v8.1*

- Developed satellite-based soil-condition screening using Sentinel-2, Landsat 8, Sentinel-1 SAR, SRTM and CHIRPS.
- Derived relative proxy indicators for bare soil, organic-matter screening, salinity, clay/mineral content and SAR-based relative soil moisture, explicitly distinguished from laboratory measurements.
- Built composite soil-quality/degradation indices and separate water- and wind-erosion screening models using weighted, direction-standardized factors; added percentile thresholds, classified condition/risk categories, area statistics and seeded spatial sampling.
- Separated processing from interface/reporting and prepared GeoTIFF outputs for downstream GIS workflows.

PROFESSIONAL EXPERIENCE

Computer & Laptop Technician / Sales - Al-Bakkour Technology & Computer | *Hama, Syria | 2019-2025*

- Provided hands-on diagnostics, repair and customer-facing technical support across more than 1,000 desktop computers and laptops.
- Installed and configured Windows, drivers, applications and BIOS settings; replaced/upgraded RAM, SSD/HDD, screens and batteries.
- Installed/configured printers, scanners, CCTV/DVR and POS systems; handled basic LAN/Wi-Fi setup and connectivity troubleshooting.
- Handled approximately 15-20 service cases per day across troubleshooting, sales and after-sales support.

OTHER PROJECTS

AGRI-EYE project website: agri-eye3k.pages.dev - multilingual portfolio website presenting the platform and its workflows.

KM COMPANY website: km-company.netlify.app - responsive company website and front-end project.

Dental-clinic website - independent front-end project covering service presentation, interaction and web deployment.

EDUCATION, TRAINING & LANGUAGES

Education: Secondary School Certificate - Scientific Stream, Syria | 2011

Training: Google IT Support Professional Certificate - in progress; CompTIA A+ - in progress

Languages: Arabic - Native; English - B2 (EF SET); French - basic working knowledge

Professional availability: Open to relocation and suitable opportunities in Italy, Canada and elsewhere in Europe; work authorization depends on the relevant legal route and employer.

Independent development timeline: GIS study from 2022; Google Earth Engine crop-monitoring work from 2023; desktop app, reporting and multilingual workflows expanded during 2025-2026.